Learning with Errors with Output Dependencies: LWE, LWR, and Physical Learning Problems under the Same Umbrella

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Abstract. Learning problems have become a foundational element for constructing quantum-resistant cryptographic schemes, finding broad application even beyond, such as in Fully Homomorphic Encryption. The increasing complexity of this field, marked by the rise of physical learning problems due to research into side-channel leakage and secure hardware implementations, underscores the urgent need for a more comprehensive analytical framework capable of encompassing these diverse variants.

In response, we introduce *Learning With Errors with Output Dependencies (LWE-OD)*, a novel learning problem defined by an error distribution that depends on the inner product value and therefore on the key. LWE-OD instances are remarkably versatile, generalizing both established theoretical problems like Learning With Errors (LWE) or Learning With Rounding (LWR), and emerging physical problems such as Learning With Physical Rounding (LWPR).

Our core contribution is establishing a reduction from LWE to LWE-OD. This is accomplished by leveraging an intermediate problem, denoted qLWE. Our reduction follows a two-step, simulator-based approach, yielding explicit conditions that guarantee LWE-OD is at least as computationally hard as LWE. While this theorem provides a valuable reduction, it also highlights a crucial distinction among reductions: those that allow explicit calculation of target distributions versus weaker ones with conditional results. To further demonstrate the utility of our framework, we offer new proofs for existing results, specifically the reduction from

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LWE to LWR and from LPN to Learning Parity with Noise with Output Dependencies (LPN-OD). This new reduction opens the door for a potential reduction from LWE to LWPR.

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1 Introduction

Learning problems. Learning problems provide a simple yet powerful computational abstraction: while solving exact linear systems is efficient, adding small amounts of noise makes the problem hard to invert. This hardness assumption has given rise to several foundational problems in cryptography, including Learning Parity with Noise (LPN), Learning With Errors (LWE) [Reg05], and Learning With Rounding (LWR) [BPR12]. These problems have become central tools in the design of public-key encryption, digital signatures, and fully homomorphic encryption (FHE) [Gen09], as well as in the development of post-quantum cryptography.
Among them, LPN, defined over the binary field, is the most elementary and has been widely used in lightweight protocols and authentication schemes [Pie12]. LWE, on the other hand, operates over larger moduli and benefits from reductions to worst-case lattice problems, making it the cornerstone of many lattice-based post-quantum cryptosystems.
While LPN and LWE are often presented under the unified umbrella of learning problems, it is important to note that LPN is deeply rooted in code-based cryptography, where it corresponds to the problem of decoding a random linear code. Historically, this challenge has been studied for decades, leading to early cryptographic primitives like Alekhnovitch's encryption [Ale03] which predated the formalization of LWE. Although LPN can be viewed as a specific case of LWE with a binary modulus $q = 2$, they rely on fundamentally different error constraints: LWE typically focuses on 'small' noise modulo q, whereas code-based cryptography (and LPN) deals with 'sparse' noise, i.e., a low Hamming weight. In this sense, these problems can be more broadly categorized as error-based cryptography.
Early constructions relied on unstructured versions of LWE, but for improved efficiency, recent schemes such as Crystals-Kyber, Crystals-Dilithitum and Saber [DKL+18,DKRV18] use structured variants like Module-LWE and Module-LWR. Learning problems are also at the heart of fully homomorphic encryption. Gentry's original FHE construction [Gen09] was

built on ideal lattices, and most practical schemes today, such as TFHE [CGGI20] and CKKS [CKKS17], rely on structured LWE-based assumptions.

Inexact computing. The efficient implementation of learning problems has become a critical objective due to their increasing centrality in post-quantum cryptography. Traditionally, implementations compute an inner product and subsequently add an independently generated error using a Pseudo-Random Number Generator (PRNG). However, a recent trend, known as inexact computing, has emerged from hardware design, observing that hardware operating under over-voltage or utilizing analog components can perform operations approximately. This leads to slight but systematic deviations in the output.
By leveraging and controlling these inherent imperfections, it becomes possible to directly generate faulty or "noisy" inner products, thus obviating the explicit computation of the error. Since the PRNG is often the most resource-intensive component, removing its use can significantly reduce power consumption and improve speed. The first experimental demonstration of inexact computing applied to a learning problem was presented by Kamel et al. [KSD+20]. In their work, the authors designed a prototype processor implementing the Learning Parity with Noise (LPN) problem where the hardware was affected by voltage overscaling.
This approach resulted in a functional system where the requisite noise was naturally generated by the physical behavior of the circuit, rather than being artificially introduced in software. Beyond efficiency, Kamel et al. also highlighted the benefits for passive side-channel security. In traditional implementations, the PRNG is often considered a weak point. By removing it, the computation of the noisy inner product becomes linear in the secret key, enabling the use of less expensive masking techniques to enhance security. This implementation, however, introduces extra challenges for side-channel analysis, as fault attacks may inadvertently alter the system's inherent error generation.

Output-dependent errors. A significant drawback of generating noise for learning problems via inexact computing is the introduction of nontrivial noise patterns that depend on the actual values being processed. In cryptographic schemes that rely on learning problems, this results in a key-dependent error, which immediately raises a critical question: does this systematic, non-random noise compromise the security of the scheme, and how should it be formally modeled?
Building on this concern, Bellizia et al. [BHK+21] introduced a formal framework for such implementations, coined Learning Parity with Physical Noise (LPPN). Since the underlying physical function is, by its nature, impossible to model exactly, they abstracted the circuit's behavior into a mathematical variant of LPN with an output-dependent error, which they denoted as LPN-OD. Crucially, they formally proved a tight, two-way reduction between LPN-OD and the standard LPN problem, thereby establishing the security basis for their physical approach.
Inexact computing was later applied to a different problem space with the Learning With Errors (LWE) problem in a subsequent work [BHK+22], which introduced the Learning With Physical Errors (LWPE) problem.

In the LWPE model, the error distribution in LWE samples may depend on the inner product between the sample and the secret. However, in this higher-dimensional setting, the authors addressed these output dependencies through heuristic arguments. They demonstrated that by using clever implementation choices, the amplitude of this dependency could be significantly reduced. While the resulting dependencies in the LWPE scheme appeared small enough not to pose an immediate security threat, a formal security reduction from standard LWE to a mathematical model of LWPE was still needed to provide a rigorous guarantee of the implementation's security. This important step remained an open problem.

Learning with errors with output-dependent noise The difficulties encountered when modeling physical or inexact implementations of LWE highlight the need for a more flexible theoretical framework. In such scenarios, the noise distribution is not fixed but may vary depending on the value of the inner product $\langle a, k \rangle$, due to the physical characteristics of the computation. This breaks the standard assumption in LWE that the error is sampled independently from a global distribution and suggests that a richer model is required to accurately capture the behavior of these systems.

To address this, we introduce the Learning With Errors with Output Dependencies (LWE-OD) problem. This model generalizes LWE by allowing the error distribution to depend explicitly on the inner product value. More precisely, given a modulus q, the LWE-OD distribution is defined by a family of q error distributions $(\varphi_0, \ldots, \varphi_{q-1})$ over $\mathbb{F}_q$. A sample is generated by selecting a uniformly random vector $a \in \mathbb{Z}_q^n$, computing the inner product $i = \langle a, k \rangle$, and adding an error drawn from φ_i. The resulting sample takes the form $(a, \langle a, k \rangle + e)$, where $e \leftarrow \varphi_{\langle a,k \rangle}$.

This construction preserves the core structure of LWE while introducing the flexibility to model data-dependent or implementation-specific noise. The LWE-OD formulation is particularly natural in the context of physical learning problems, where such dependencies arise from analog behavior, reduced precision, or leakage, and can slightly vary from one implementation to another. By formalizing these variations within a single model, LWE-OD provides a starting point for understanding the security of non-ideal or physically grounded LWE instantiations.

Broader applications. The Learning With Errors with Output-Dependent Noise (LWE-OD) framework, while initially motivated by the need to model inexact computing and physically induced noise, possesses an applicability that extends far beyond this specific setting. It offers a natural, unified abstraction to capture a broad class of learning problems that share the essential algebraic structure of LWE but exhibit more complex or context-specific noise behaviors.

The LWE-OD framework is highly versatile, as it encompasses several classical learning problems as special cases:

- Standard LWE: The original LWE setting is recovered if all error distributions $\varphi_0, \ldots, \varphi_{q-1}$ are identical (i.e., independent of the inner product value). As LPN is a specific instance of LWE, it too falls under the generalization provided by LWE-OD.

- Learning With Rounding (LWR): Introduced by Banerjee et al. in [BPR12], this important variant is defined by a deterministic rounding function applied to the secret inner product. This problem is modeled when the distributions φ_i are deterministic and simulate quantization or truncation of the inner product.
- LPN-OD is the particular case where the modulus is $q = 2$.

LWE-OD also accommodates more exotic variants. A particularly relevant example is the Learning With Physical Rounding (LWPR) problem, introduced by Duval et al. [DMMS21] and extended by Hoffmann et al. [HMM+23]. LWPR captures scenarios where the inner product $\langle \mathbf{a}, \mathbf{k} \rangle$ is not directly available, but instead leaks through a rounding function applied during physical computation. LWPR was initially introduced as an efficient rekeying block for symmetric cryptography. However, this type of problem is also useful for analyzing side-channel attacks on post-quantum schemes [HLM+23], or in the context of Fully Homomorphic Encryption (FHE), where rounding is used to control noise growth. In these cases, the deterministic, data-dependent leakage introduced by rounding fits seamlessly within the LWE-OD abstraction.

These variations highlight the flexibility of learning problems as a design tool, but they also complicate security analyses. In particular, they motivate the development of more general frameworks capable of encompassing practical, structured, or physically motivated deviations from the original problems. The LWE-OD model answers this need by providing a unified abstraction that accommodates both classical problems and implementation-driven variants, while preserving the essential structure needed for security analysis.

Contributions. Our primary contribution is the formal introduction of this novel learning problem: **Learning With Error with Output Dependencies (LWE-OD)** in Definition 5, that encompasses many known problems as detailed in Proposition 1.

We then dedicate the major portion of our work to establishing a reduction from LWE to LWE-OD. Theorem 3 provides explicit conditions on the error distributions of LWE-OD and LWE instances that, if met, guarantee the LWE-OD instance is at least as computationally hard as the LWE one.

This reduction is realized in two steps, with the introduction of an intermediary problem denoted qLWE. A qLWE sample is composed of q different LWE samples, under the same key but using (possibly) different error distributions. This problem is used as an intermediate step for the full reduction, Theorem 1 providing the reduction from qLWE to LWE-OD. This first reduction leverages a simulator employing rejection sampling, demonstrating that an adversary can transform LWE-OD samples into qLWE samples (under the same key) in polynomial time, thereby leading to a tight reduction. It ensures that, for any LWE-OD instance, it is always possible to explicitly derive a corresponding qLWE instance such that LWE-OD $\geq$ qLWE.

Subsequently, Theorem 2 presents a reduction from LWE to qLWE. This second stage also relies on a simulator, proving that strategically adding well-chosen error to LWE samples can effectively simulate a qLWE sample. These two reductions are then combined in Theorem 3 to form the complete reduction from LWE to LWE-OD.

While our main theorem provides a valuable reduction from LWE to LWE-OD, we emphasize that the combined reduction in Theorem 3 is subject to strong limitations. Unlike the intermediate steps, the final reduction is not always constructive in a general sense.

The reduction relies on explicit conditions involving the Walsh transforms of the error distributions to ensure they remain valid probability distributions after deconvolution. In practice, these conditions constitute a significant constraint:

- Non-invertibility: Given a standard LWE instance, it is not possible in the general case to derive a non-trivial family of LWE-OD instances that are guaranteed to be at least as hard as the LWE instance.
- Parameter Derivation: For a given LWE-OD instance, it is not immediately clear how to identify the optimal LWE distribution (the one minimizing the LWE advantage) from which it reduces.

As illustrated in Figure 1, our framework primarily serves as a **verification tool**. It allows for checking a reduction between two existing problem instances rather than providing a universal method to instantiate arbitrary LWE-OD problems from any LWE source.

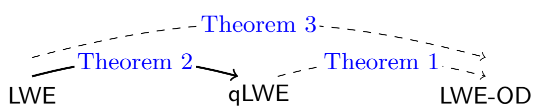

Fig. 1. Reductions between various learning problems explored in this work. Plain directed arrows signify reductions with explicit target distribution, while dashed arrows denote reductions that provide verification conditions.

Finally, we demonstrate the versatility of our main theorem by applying it to different problem variants, yielding both new and previously established security proofs:

- New proofs for known results: Our framework provides a novel proof for the reduction from LWE to LWR originally by Bogdanov et al. [BGM+16], as well as a new proof for the reduction from LPN to LPN-OD by Bellizia et al. [BHK+21].
- Novel extensions: We extend the LWE to LWR reduction to the case where the rounding modulus p does not divide the ring modulus q.

We believe that our LWE-OD framework and the associated reductions will serve as foundational tools for providing assurance to inexact computing in practical implementations. By providing a unified approach to handle output-dependent noise, we hope to open new avenues for both theoretical exploration and practical application of learning-based cryptography.

2 Notations and preliminaries

For the rest of the paper, $n \in \mathbb{N}$ is a dimension parameter, $q \in \mathbb{N}$ is a modulus polynomial in n. The notation $\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$ denotes the ring modulus q. When applied to elements of $\mathbb{Z}_q$, the symbol $+$ denotes addition modulo q. This notation appears frequently, though not exclusively, within summation indices. $\mathbb{Z}_q$ distributions refer to probability distributions over $\mathbb{Z}_q$. To ensure a uniform distribution of $\langle \boldsymbol{a}, \boldsymbol{k} \rangle$ for a uniform $\boldsymbol{a} \in \mathbb{Z}_q^n$, we define $(\mathbb{Z}_q)^*$ the set of invertible elements of $\mathbb{Z}_q$ (that will be the definition set of the elements of $\boldsymbol{k}$). We denote by $(\mathbb{Z}_q^n)^*$ the set of elements in $\mathbb{Z}_q^n$ such as at least one of the coefficient is invertible. Let φ be a $\mathbb{Z}_q$-distribution, $X \leftarrow \varphi$, $\varphi(i)$ denotes the probability $\Pr[X = i]$. For φ, χ $\mathbb{Z}_q$-distributions, and X, Y independent random variables such that $X \leftarrow \varphi$, and $Y \leftarrow \chi$, we denote by $\varphi * \chi$ the $\mathbb{F}_q$-distribution followed by $X + Y$. Let Ber denote a Bernouilli distribution.

Let us introduce the Learning with Errors (LWE) distribution, first introduced in [Reg05], which is a fundamental problem in lattice-based cryptography. It is defined as follows.

Definition 1 (The LWE distribution). *Let $n \in \mathbb{N}^*, q \in \mathbb{Z}$ be parameters. Let $\boldsymbol{k} \in (\mathbb{Z}_q^n)^*$ be a fixed secret vector. Let ψ be a probability distributions over $\mathbb{Z}_q$. We denote by $\mathsf{LWE}_\psi^n(\boldsymbol{k})$ distribution the probability distribution over $\mathbb{Z}_q^n \times \mathbb{Z}_q$ obtained from Sampler 1.*

Sampler 1: $\mathsf{LWE}_\psi^n(\boldsymbol{k})$

$\boldsymbol{a} \leftarrow \mathbb{Z}_q^n$
$\boldsymbol{e} \leftarrow \psi$
$b := \langle \boldsymbol{a}, \boldsymbol{k} \rangle + e \in \mathbb{Z}_q$
Return $(\boldsymbol{a}, b)$

In this setting, the secret key $\boldsymbol{k}$ is an arbitrary vector in $(\mathbb{Z}_q^n)^*$, without any assumption on its distribution. The only requirement is that $\boldsymbol{k}$ is in $(\mathbb{Z}_q^n)^*$ and not simply in $\mathbb{Z}_q^n$, which ensures that $\langle \boldsymbol{a}, \boldsymbol{k} \rangle$ is uniformly distributed when $\boldsymbol{a}$ is uniform.
We also introduce the Learning with rounding (LWR) distribution, first introduced in [BPR12], which is an alternative lattice-based problem that is closely related to the LWE problem. It has the advantage of being less complex to sample from, as it does not require a specific error distribution. It is defined as follows.

Definition 2 (The LWR distribution). *Let $n, p \leq q \in \mathbb{N}^*$ be parameters. Let $\boldsymbol{k} \in (\mathbb{Z}_q^n)^*$ be a fixed secret vector. Let* round *be a rounding function such that* $\mathsf{round}_p(x) = \lfloor (p/q) \cdot x \rceil$ *for $x \in \mathbb{Z}_q$. We denote by* $\mathsf{LWR}_{p,q}^n(\boldsymbol{k})$ *distribution the probability distribution over $\mathbb{Z}_q^n \times \mathbb{Z}_p$ obtained from Sampler 2.*

Sampler 2: $\mathsf{LWR}^n_{p,q}(\boldsymbol{k})$

$\boldsymbol{a} \leftarrow \mathbb{Z}_q^n$
$b := \mathsf{round}_p(\langle \boldsymbol{a}, \boldsymbol{k} \rangle)$
Return $(\boldsymbol{a}, b)$

The next definition introduces the Learning Parity with Noise (LPN) distribution, which is a fundamental problem in cryptography and coding theory, first introduced in [BKW00]. It is an oracle-based problem consisting of decoding a random code whose rate can be chosen arbitrarily close to 0. This problem has been widely studied in the literature and enjoys stong security guarantees such as worst to average-case reductions [BLVW19]. It allows to build authentication protocols [HB01], one way-functions [BFKL94], secret-key encryption schemes [GRS08], public-key encryption schemes [DV13], pseudorandom functions [YS16] or hash functions [YZW$^+$19]. We present its definition below.

Definition 3 (The LPN distribution). *Let $n \in \mathbb{N}^*$ be an integer and $\tau \in \mathbb{R}$, $0 < \tau < 0.5$ be a noise rate. Let $\boldsymbol{k} \in (\mathbb{F}_2^n)^*$ be a fixed secret vector. We denote by $\mathsf{LPN}^n_\tau(\boldsymbol{k})$ distribution the probability distribution over $\mathbb{F}_2^n \times \mathbb{F}_2$ obtained from Sampler 3.*

Sampler 3: $\mathsf{LPN}^n_\tau(\boldsymbol{k})$

$\boldsymbol{a} \leftarrow \mathbb{F}_2^n$
$\boldsymbol{e} \leftarrow \mathsf{Ber}(\tau)$
$b := \langle \boldsymbol{a}, \boldsymbol{k} \rangle + e \in \mathbb{F}_2$
Return $(\boldsymbol{a}, b)$

It is important to note that the presented instantitations of LWE, LWR and LPN are actually abstractions. Indeed, in practice, the number of samples is fixed in advance and not polynomial. This limitation could lead to different attack regimes outside of the scope of this paper.

Kamel et al. [KSD$^+$20] introduces Learning Parity with Physical Noise (LPPN), a physical variant of the LPN problem. This definition was motivated by the use of inexact computing to achieve more efficient and secure hardware implementations. In this variant, the error distribution stems from physical miscomputations occurring during the computation of $\langle \boldsymbol{a}, \boldsymbol{k} \rangle$, where $\boldsymbol{a}$ is a uniformly random vector in $\mathbb{Z}_2^n$ and $\boldsymbol{k}$ is the secret key. As a result, the error distribution is inherently dependent on the secret key $\boldsymbol{k}$. Since this dependency is given by a physical function, the problem is difficult to analyze directly. This difficulty led Bellizia et al. to introduce the LPN-OD distribution ([BHK+21], described below), which serves as a mathematical approximation of the LPPN problem.

Definition 4 (The LPN-OD distribution [BHK+21]). *Let $n \in \mathbb{N}^*$ be an integer. Let $\boldsymbol{k} \in (\mathbb{F}_2^n)^*$ be a fixed secret vector. Let φ_0, φ_1 be two probability distributions over $\mathbb{F}_2$. We denote by $\mathsf{LPN\text{-}OD}^n_{\varphi_0,\varphi_1}(\boldsymbol{k})$ distribution the probability distribution over $\mathbb{F}_2^n \times \mathbb{F}_2$ obtained from Sampler 4.*

Sampler 4: $\mathsf{LPN\text{-}OD}^n_{\varphi_0,\varphi_1}(\boldsymbol{k})$

$\boldsymbol{a} \leftarrow \mathbb{F}_2^n$
$i := \langle \boldsymbol{a}, \boldsymbol{k} \rangle \in \mathbb{F}_2$
$\boldsymbol{e} \leftarrow \varphi_i$
$b := \langle \boldsymbol{a}, \boldsymbol{k} \rangle + e \in \mathbb{F}_2$
Return $(\boldsymbol{a}, b)$

LPPN was subsequently generalized to higher modulus as the LWPE problem in [BHK+22]. Similar to its predecessor, LWPE's implementation exhibits physical key dependencies. Current work [BHK+22] addresses these dependencies heuristically through specific implementation choices, leaving the theoretical study of LWPE as an open problem. This work addresses that open problem by first defining a mathematical approximation of LWPE.

3 Learning with errors and with output dependencies

In this section, similar to the introduction of LPN-OD, we naturally extend the LWE distribution to the case where the error distribution depends on the secret key. This generalization is also originally motivated by the study of inexact computing. However, this generalization extends beyond the inexact computing setting and encompasses a broader class of distributions, such as LWR. We now formalize this notion of output dependency in the following definition.

Definition 5 (The LWE-OD distribution). *Let $n, q \in \mathbb{N}^*$ be parameters. Let $\boldsymbol{k} \in (\mathbb{Z}_q^n)^*$ be a fixed secret vector. Let $\varphi_0, \ldots, \varphi_{q-1}$ be a collection of q probability distributions over $\mathbb{Z}_q$. We denote by $\mathsf{LWE\text{-}OD}^n_{\varphi_0,\ldots,\varphi_{q-1}}(\boldsymbol{k})$ distribution the probability distribution over $\mathbb{Z}_q^n \times \mathbb{Z}_q$ obtained from Sampler 5.*

Sampler 5: $\mathsf{LWE\text{-}OD}^n_{\varphi_0,\ldots,\varphi_{q-1}}(\boldsymbol{k})$

$\boldsymbol{a} \leftarrow \mathbb{Z}_q^n$
$i := \langle \boldsymbol{a}, \boldsymbol{k} \rangle \in \mathbb{Z}_q$
$\boldsymbol{e} \leftarrow \varphi_i$
$b := \langle \boldsymbol{a}, \boldsymbol{k} \rangle + e$
Return $(\boldsymbol{a}, b)$

One can naturally introduce the search problems associated with all the introduced distributions. The search problem consists in finding the secret key $\boldsymbol{k}$ given access to a sampler for the distribution. The search problem is said to be hard if no efficient algorithm can solve it with a non-negligible probability.
The interest of LWE-OD is the fact that it generically includes all previously introduced distributions. The following proposition summarizes the relationships between the LWE-OD distribution and the previously introduced distributions. Its proof is straightforward as it consists in applying the definitions.

Proposition 1. *The following distributions are particular cases of the* LWE-OD *distribution:*
- *The Learning with Errors (*LWE*) problem, where $\varphi_i = \psi$ for all i.*
- *The Learning with Rounding (*LWR*) problem, where φ_i is the distribution that outputs $\mathsf{round}_p(i) - i$ with probability 1.*
- *The Learning Parity with Noise (LPN) problem, $q = 2$ and $\varphi_i = \mathsf{Ber}_\tau$.*
- *The Learning Parity with Noise with Output Dependencies (LPN-OD) problem, where $q = 2$.*

A generic reduction between LWE-OD and LWE seems for now inachievable because of the large spectrum of possibilities for LWE-OD. Any conditional reduction is still an important achievement in the field of lattice-based cryptography, as it allows to leverage the security of the LWE problem to the more general case of LWE with output dependencies, that is relevant in many scenarios. This paper proposes a reduction in the next section.

4 Towards a reduction between LWE-OD to LWE

In this section, we present the main contribution of this paper. We provide a reduction from the LWE-OD search problem to the LWE search problem.
This section is structured as follows: we first introduce the qLWE distribution, next we provide a complete reduction from the qLWE distribution to the LWE-OD distribution, and we show how to to reduce from the qLWE search problem to the LWE search problem under some conditions.

4.1 An intermediate LWE variant

Let us introduce the qLWE distribution. Intuitively, this distribution is the combination of q LWE ones with (potentially) different error distributions but under the same key. $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ is at most as secure as the weakest $\mathsf{LWE}^n_{\chi_i}(\boldsymbol{k})$, and is only used as a milestone towards a reduction from LWE.

Definition 6 (qLWE distribution).
Let $n, q \in \mathbb{N}^*$ be parameters. Let $\boldsymbol{k} \in (\mathbb{Z}_q^n)^*$ be a fixed secret vector. Let $\chi_0, \ldots, \chi_{q-1}$ be a collection of q probability distributions over $\mathbb{Z}_q$. We denote by $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ distribution the probability distribution over $\left(\mathbb{Z}_q^n \times \mathbb{Z}_q\right)^q$ obtained from Sampler 6.

Sampler 6: $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$

For $i \in [0, q-1]$:
$\quad \boldsymbol{a}_i \leftarrow \mathbb{Z}_q^n$
$\quad \boldsymbol{e}_i \leftarrow \chi_i$
$\quad b_i := \langle \boldsymbol{a_i}, \boldsymbol{k} \rangle + e_i$
Return $(\boldsymbol{a}_i, b_i)_{0 \le i < q}$

Note that one can also straightforwardly introduce the qLWE problem as the search problem associated with the $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ distribution. Note that this definition and thus the reduction presented in the next section are only valid if q is polynomial in n. Considering exponential q is left as a future analysis.

4.2 Step 1: a reduction from qLWE to $\mathsf{LWE\text{-}OD}$

In this section, we give a condition under which the $\mathsf{LWE\text{-}OD}$ search problem is at least as hard as the qLWE search problem. After giving a characterization of the $\mathsf{LWE\text{-}OD}^n_{\varphi_0,\ldots,\varphi_{q-1}}(\boldsymbol{k})$ distribution, we show that it is possible to create $\mathsf{LWE\text{-}OD}^n_{\varphi_0,\ldots,\varphi_{q-1}}(\boldsymbol{k})$ samples using a $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ oracle. We finally use this sampler to give a security reduction between $\mathsf{LWE\text{-}OD}$ and qLWE.

$$\mathsf{qLWE} \xrightarrow{\text{Theorem 1}} \mathsf{LWE\text{-}OD}$$

Firstly, we give a technical characterization of the $\mathsf{LWE\text{-}OD}$ samples.
To avoid overloading notations, we extend $\varphi_i(j)$ to all integers $i, j \in \mathbb{Z}$ by periodicity: $\varphi_{i+kq} = \varphi_i$ and $\varphi_i(j + kq) = \varphi_i(j)$ for all integers k and indices $i, j \in [0, q-1]$.

Proposition 2 ($\mathsf{LWE\text{-}OD}^n_{\varphi_0,\dots,\varphi_{q-1}}(\boldsymbol{k})$ distribution characterization). *Let $n, q \in \mathbb{N}^*$ be parameters. Let $\boldsymbol{k} \in (\mathbb{Z}_q^n)^*$ be a fixed secret vector. Let $\varphi_0, \dots, \varphi_{q-1}$ be a collection of q probability distributions over $\mathbb{Z}_q$. The distribution of $\mathsf{LWE\text{-}OD}^n_{\varphi_0,\dots,\varphi_{q-1}}(\boldsymbol{k})$ over $\mathbb{Z}_q^n \times \mathbb{Z}_q$ is characterized as follows. For any $\boldsymbol{l} \in \mathbb{Z}_q^n$ and $j \in \mathbb{Z}_q$,*

$$Pr[\boldsymbol{a} = \boldsymbol{l}, b = j] \quad = \varphi_{\langle \boldsymbol{l}, \boldsymbol{k} \rangle}(j - \langle \boldsymbol{l}, \boldsymbol{k} \rangle) \cdot \frac{1}{q^n}.$$

Proof. Let $n, q \in \mathbb{N}^*$ $\boldsymbol{k} \in (\mathbb{Z}_q^n)^*$ and $\varphi_0, \dots, \varphi_{q-1}$ be as stated in the proposition.
For any $\boldsymbol{l} \in \mathbb{Z}_q^n$ and $j \in \mathbb{Z}_q$,

$$\begin{aligned}
&\Pr[\boldsymbol{a} = \boldsymbol{l}, b = j] \\
&= \Pr[b = j | \boldsymbol{a} = \boldsymbol{l}] \cdot \Pr[\boldsymbol{a} = \boldsymbol{l}] && \text{(Definition of conditional probability)} \\
&= \Pr[b = j | \langle \boldsymbol{a}, \boldsymbol{k} \rangle = \langle \boldsymbol{l}, \boldsymbol{k} \rangle] \cdot \Pr[\boldsymbol{a} = \boldsymbol{l}] && \text{(As } b \text{ only depends on } \langle \boldsymbol{a}, \boldsymbol{k} \rangle\text{)} \\
&= \Pr[\langle \boldsymbol{a}, \boldsymbol{k} \rangle + e = j | \langle \boldsymbol{a}, \boldsymbol{k} \rangle = \langle \boldsymbol{l}, \boldsymbol{k} \rangle] \cdot \Pr[\boldsymbol{a} = \boldsymbol{l}] \\
&= \Pr[\langle \boldsymbol{a}, \boldsymbol{k} \rangle + e = j | \langle \boldsymbol{a}, \boldsymbol{k} \rangle = \langle \boldsymbol{l}, \boldsymbol{k} \rangle] \cdot \frac{1}{q^n} && \text{(As by definition } \boldsymbol{a} \leftarrow \mathbb{Z}_q^n\text{)} \\
&= \Pr[e = j - \langle \boldsymbol{l}, \boldsymbol{k} \rangle | \langle \boldsymbol{a}, \boldsymbol{k} \rangle = \langle \boldsymbol{l}, \boldsymbol{k} \rangle] \cdot \frac{1}{q^n} \\
&= \varphi_{\langle \boldsymbol{l}, \boldsymbol{k} \rangle}(j - \langle \boldsymbol{l}, \boldsymbol{k} \rangle) \cdot \frac{1}{q^n}.
\end{aligned}$$

This concludes the proof of the proposition.
As a sanity check, let us show that the probabilities sum to 1.

$$\sum_{\boldsymbol{l} \in \mathbb{Z}_q^n} \sum_{j \in \mathbb{Z}_q} \Pr[\boldsymbol{a} = \boldsymbol{l}, b = j] = \sum_{\boldsymbol{l} \in \mathbb{Z}_q^n} \sum_{j \in \mathbb{Z}_q} \varphi_{\langle \boldsymbol{l}, \boldsymbol{k} \rangle}(j - \langle \boldsymbol{l}, \boldsymbol{k} \rangle) \cdot \frac{1}{q^n} \quad = \sum_{\boldsymbol{l} \in \mathbb{Z}_q^n} \frac{1}{q^n} = 1.$$

□

We introduce an alternative sampler and we later provide the conditions under which its output follows a $\mathsf{LWE\text{-}OD}$ distribution.

Proposition 3 (Algorithm 1 termination and complexity analysis). *Let $n \in \mathbb{N}^*$ and $q \in \mathbb{N}^*$ of polynomial size in n. Let $m \in \mathbb{N}^*$ polynomial large enough to grow at least proportionally to $q \cdot \log(n)$. Assume an access to a $\mathsf{qLWE}^n_{\chi_0,\dots,\chi_{q-1}}(\boldsymbol{k})$ sampler with fixed distributions $\chi_0, \dots, \chi_{q-1}$ and a key $\boldsymbol{k} \in (\mathbb{Z}_q^n)^*$.*
*Then, Algorithm 1 with inputs q, m, and $\mathsf{qLWE}^n_{\chi_0,\dots,\chi_{q-1}}(\boldsymbol{k})$ terminates the **while** loop with probability $1/q$ in each iteration, runs in polynomial time in n, and returns $\perp$ with probability negligible in n.*

Proof. First, since the number of calls to Sampler 6 is bounded by m and every other operation runs in constant time, for m polynomial in n the execution time is polynomial in n.
Let us next compute the acceptance probability in Algorithm 1.

Algorithm 1: AltSampleLWE-OD

Input: A sampling upper limit m, a $\mathbb{Z}_q$-distribution ρ, a $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ sampler

Output: A sample $(\boldsymbol{a}, b) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$.

$b_0 \leftarrow \rho$;
cnt $:= 0$;
do
    $(\boldsymbol{a}, b_1) \leftarrow$ Sample-$\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})[b_0]$;
    `// Only keep the` b_0`-th sample, which follows a` $\mathsf{LWE}^n_{\chi_{b_0}}(\boldsymbol{k})$ `distribution`
    cnt $\leftarrow$ cnt $+1$;
while $b_0 \neq b_1$ and $cnt < m$;
if cnt $= m$ **then**
    return $\perp$
end
return $(\boldsymbol{a}, b_1)$

Denoting by X the number of iterations of this loop and since each call to Sampler 6 is independent, X follows a geometric law of parameter $\frac{1}{q}$. Then, $\Pr[\perp] = \Pr[X > m] = \left(\frac{q-1}{q}\right)^m$, therefore negligible because m is large enough to grow at least proportionally to $q \cdot \log(n)$.

$\square$

Proposition 4 (Algorithm 1 output characterization). *Let $(\boldsymbol{a}, b) = (\boldsymbol{a}, \langle \boldsymbol{a}, \boldsymbol{k} \rangle + e)$ be a non $\perp$ output of Algorithm 1 for inputs m, ρ, and $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ for a fixed key $\boldsymbol{k} \in (\mathbb{Z}_q^n)^*$ and let $\chi_0, \ldots, \chi_{q-1}$ be distributions. The distribution of $(\boldsymbol{a}, b)$ is such that for any $j \in \mathbb{Z}_q$ and $\boldsymbol{l} \in \mathbb{Z}_q^n$,*

$$Pr[\boldsymbol{a} = \boldsymbol{l}, b = j] = \frac{1}{q^{n-1}} \cdot \chi_j(j - \langle \boldsymbol{l}, \boldsymbol{k} \rangle) \cdot \rho(j).$$

Proof. Let $\boldsymbol{k} \in (\mathbb{Z}_q^n)^*$ be the key, and $(\boldsymbol{a}^{\text{out}}, b^{\text{out}}) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$ be the output of Algorithm 1 (we assume that the output is not $\perp$).

$$
\begin{aligned}
&\Pr\left[\boldsymbol{a}^{\text{out}} = \boldsymbol{l}, b^{\text{out}} = j\right] \\
&= \frac{\Pr\left[\boldsymbol{a} = \boldsymbol{l}, b_1 = j\right]}{\Pr\left[b_1 - b_0 = 0\right]} \quad (\boldsymbol{a} \text{ and } b_1 \text{ are the values drawn inside the loop}) \\
&= \frac{\Pr\left[\boldsymbol{a} = \boldsymbol{l}, b_1 = j\right]}{\frac{1}{q}} \quad (\text{Probability of going out of the while loop}) \\
&= q \cdot \Pr\left[\boldsymbol{a} = \boldsymbol{l}, b_1 = j | b_0 = j\right] \cdot \Pr\left[b_0 = j\right] \quad (\text{Bayes formula}) \\
&= q \cdot \Pr\left[\boldsymbol{a} = \boldsymbol{l}, b_1 = j | b_0 = j\right] \cdot \rho(j) \\
&= q \cdot \Pr\left[\boldsymbol{a} = \boldsymbol{l}, \langle \boldsymbol{a}, \boldsymbol{k} \rangle + e = j | b_0 = j\right] \cdot \rho(j) \\
&= q \cdot \Pr\left[\langle \boldsymbol{a}, \boldsymbol{k} \rangle + e = j | \boldsymbol{a} = \boldsymbol{l}, b_0 = j\right] \cdot \Pr\left[\boldsymbol{a} = \boldsymbol{l} | b_0 = j\right] \cdot \rho(j) \\
&= q \cdot \Pr\left[e = j - \langle \boldsymbol{l}, \boldsymbol{k} \rangle | b_0 = j\right] \cdot \Pr\left[\boldsymbol{a} = \boldsymbol{l} | b_0 = j\right] \cdot \rho(j) \\
&= q \cdot \chi_j(j - \langle \boldsymbol{l}, \boldsymbol{k} \rangle) \cdot \Pr\left[\boldsymbol{a} = \boldsymbol{l} | b_0 = j\right] \cdot \rho(j) \\
&= q \cdot \chi_j(j - \langle \boldsymbol{l}, \boldsymbol{k} \rangle) \cdot \frac{1}{q^n} \cdot \rho(j) \quad (\text{As } \boldsymbol{a} \text{ is uniform within an iteration}) \\
&= \frac{1}{q^{n-1}} \cdot \chi_j(j - \langle \boldsymbol{l}, \boldsymbol{k} \rangle) \cdot \rho(j).
\end{aligned}
$$

This concludes the proof of the proposition.
As a sanity check, let us show that the total probability sums to 1:

$$
\begin{aligned}
\sum_{\boldsymbol{l} \in \mathbb{Z}_q^n} \sum_{j \in \mathbb{Z}_q} \Pr\left[\boldsymbol{a}^{\text{out}} = \boldsymbol{l}, b^{\text{out}} = j\right] &= \sum_{\boldsymbol{l} \in \mathbb{Z}_q^n} \sum_{j \in \mathbb{Z}_q} \frac{1}{q^{n-1}} \cdot \chi_j(j - \langle \boldsymbol{l}, \boldsymbol{k} \rangle) \cdot \rho(j) \\
&= \frac{1}{q^{n-1}} \cdot \sum_{j \in \mathbb{Z}_q} \rho(j) \sum_{\boldsymbol{l} \in \mathbb{Z}_q^n} \chi_j(j - \langle \boldsymbol{l}, \boldsymbol{k} \rangle). \\
&= \frac{1}{q^{n-1}} \cdot \sum_{j \in \mathbb{Z}_q} \rho(j) \sum_{x \in \mathbb{Z}_q^n} \chi_j(j - x) \cdot \left| \{\boldsymbol{l} \in \mathbb{Z}_q^n | \langle \boldsymbol{l}, \boldsymbol{k} \rangle = x \bmod q\} \right|.
\end{aligned}
$$

Let us compute the cardinal $\left|\{\boldsymbol{l} \in \mathbb{Z}_q^n | \langle \boldsymbol{l}, \boldsymbol{k} \rangle = x \bmod q\}\right|$. The linear form $\phi : \mathbb{Z}_q^n \to \mathbb{Z}_q$ defined by $\phi(\boldsymbol{l}) = \langle \boldsymbol{l}, \boldsymbol{k} \rangle \bmod q$ is a group homomorphism.
Surjectivity: Because $\boldsymbol{k} \in \left(\mathbb{Z}_q^n\right)^*$, there exists at least one index i such that $k_i \in \mathbb{Z}_q^*$. Let x be an element of $\mathbb{Z}_q$, we define $\boldsymbol{l} \in \mathbb{Z}_q^n$ such that $l_i = k_i^{-1} \cdot x$ and $l_j = 0$ for all $j \neq i$, it implies $\langle \boldsymbol{l}, \boldsymbol{k} \rangle = x$. Therefore $x \in \text{Im}(\phi)$. Hence $\text{Im}(\phi) = \mathbb{Z}_q$.
Pre-image cardinality: We have $\left|\{\boldsymbol{l} \in \mathbb{Z}_q^n | \langle \boldsymbol{l}, \boldsymbol{k} \rangle = x \bmod q\}\right| = |\phi^{-1}(x)|$. For all $x \in \mathbb{Z}_q$, by the first isomorphism theorem, $|\phi^{-1}(x)| = |\ker(\phi)| = \frac{q^n}{|\text{Im}(\phi)|} = \frac{q^n}{q} = q^{n-1}$.

Continuing the computation:

$$
\begin{aligned}
\sum_{\boldsymbol{l} \in \mathbb{Z}_q^n} \sum_{j \in \mathbb{Z}_q} \Pr\left[\boldsymbol{a}^{\text{out}} = \boldsymbol{l}, b^{\text{out}} = j\right] &= \frac{1}{q^{n-1}} \cdot \sum_{j \in \mathbb{Z}_q} \rho(j) \sum_{x \in \mathbb{Z}_q} \chi_j(j - x) \cdot \left| \{\boldsymbol{l} \in \mathbb{Z}_q^n | \langle \boldsymbol{l}, \boldsymbol{k} \rangle = x \bmod q\} \right| \\
&= \frac{1}{q^{n-1}} \cdot \sum_{j \in \mathbb{Z}_q} \rho(j) \sum_{x \in \mathbb{Z}_q} \chi_j(j - x) \cdot q^{n-1} \\
&= \sum_{j \in \mathbb{Z}_q} \rho(j) \\
&= 1.
\end{aligned}
$$

□

Let us look at particular cases.

$$\begin{aligned}
\Pr\left[b^{\text{out}} = j\right] &= \sum_{\boldsymbol{l}\in\mathbb{Z}_q^n} \Pr\left[\boldsymbol{a}^{\text{out}} = \boldsymbol{l}, b^{\text{out}} = j\right] \\
&= \sum_{\boldsymbol{l}\in\mathbb{Z}_q^n} \frac{1}{q^{n-1}} \cdot \chi_j(j - \langle \boldsymbol{l}, \boldsymbol{k}\rangle) \cdot \rho(j) \\
&= \frac{q^{n-1}\cdot 1}{q^{n-1}} \cdot \rho(j) \quad \text{(using the cardinal computed above)} \\
&= \rho(j). \\
\Pr\left[a^{\text{out}} = \boldsymbol{l}\right] &= \sum_{j\in\mathbb{Z}_q} \Pr\left[\boldsymbol{a}^{\text{out}} = \boldsymbol{l}, b^{\text{out}} = j\right] \\
&= \sum_{j\in\mathbb{Z}_q} \frac{1}{q^{n-1}} \cdot \chi_j(j - \langle \boldsymbol{l}, \boldsymbol{k}\rangle) \cdot \rho(j) \\
&= \frac{1}{q^{n-1}} \cdot \sum_{j\in\mathbb{Z}_q} \chi_j(j - \langle \boldsymbol{l}, \boldsymbol{k}\rangle) \cdot \rho(j).
\end{aligned}$$

We note that $\boldsymbol{a}$ is not uniform in $\mathbb{Z}_q^n$ in general.

We now provide the main theorem of this part, which gives a reduction of $\mathsf{LWE\text{-}OD}$ from qLWE for given parameters. The high-level idea of its proof is showing that an adversary can use Algorithm 1 to turn qLWE samples into $\mathsf{LWE\text{-}OD}$ ones in polynomial time. With this transformation possible, the existence of an attack against $\mathsf{LWE\text{-}OD}$ would enable an attack against qLWE.

Theorem 1 ($\mathsf{LWE\text{-}OD}$ reduction from qLWE). *Let $n, q \in \mathbb{N}^*$ be parameters. Let $\varphi_0, \ldots, \varphi_{q-1}$ be a collection of q probability distributions over $\mathbb{Z}_q$. We define the following distributions:*

$$\forall i \in [0, q-1] \quad \rho(i) := \frac{1}{q}\sum_{k=0}^{q-1} \varphi_k(i-k)$$

and for any $j \in [0, q-1]$ such that $\rho(j) \neq 0$, let χ_j be the distribution over $\mathbb{Z}_q$ defined as:

$$\forall i \in [0, q-1] \quad \chi_j(i) := \frac{\varphi_{j-i}(i)}{q\cdot\rho(j)} = \frac{\varphi_{j-i}(i)}{\sum_{k=0}^{q-1}\varphi_k(j-k)}.$$

Then, for any adversary $\mathcal{A}$ against the $\mathsf{LWE\text{-}OD}^n_{\varphi_0,\ldots,\varphi_{q-1}}(\boldsymbol{k})$search problem, there exists an adversary $\mathcal{B}$ against the $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$search problem such that

$$\boxed{\begin{aligned}
&\Pr[\boldsymbol{k} \leftarrow (\mathbb{Z}_q^n)^* : \mathcal{A}^{\mathsf{LWE\text{-}OD}^n_{\varphi_0,\ldots,\varphi_{q-1}}(\boldsymbol{k})}(1^n) = \boldsymbol{k}] \\
\le\, & m \cdot \Pr[\boldsymbol{k} \leftarrow (\mathbb{Z}_q^n)^* : \mathcal{B}^{\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})}(1^n) = \boldsymbol{k}],
\end{aligned}}$$

where m is a polynomial large enough to grow at least proportionally to $q{\cdot}\log(n)$ and $\mathcal{A}$ and $\mathcal{B}$ both have polynomial memory and time complexity.

Remark 1. Note that if there is a $j \in [0, q-1]$ such that $\rho(j) = 0$, then χ_j can be arbitrarily defined as it will never be used in the reduction.

Proof. **A) The distributions are well defined.**
Let us first show that ρ and χ are distributions:
For ρ, since for any i, k in $[0, q-1]$ we have $\varphi_k(i) \in [0, 1]$ we get:

$$0 \leq \sum_{k=0}^{q-1} \varphi_k(i-k) \leq q,$$

hence $\rho(i) \in [0, 1]$. Then we show the total probability equals 1:

$$\sum_{i=0}^{q-1} \rho(i) = \sum_{i=0}^{q-1} \frac{1}{q} \sum_{k=0}^{q-1} \varphi_k(i-k) = \frac{1}{q} \sum_{k=0}^{q-1} \sum_{j=0}^{q-1} \varphi_k(j) = 1.$$

For χ_j, we have that for all $i \in [0, q-1]$:

$$\chi_j(i) = \frac{\varphi_{j-i}(i)}{q \cdot \rho(j)} = \frac{\varphi_{j-i}(i)}{\varphi_{j-i}(i) + \sum\limits_{[0,q-1]\setminus\{j-i\}} \varphi_k(i-k)} \leq 1,$$

and $\chi_j(i) \geq 0$ because both numerator and denominator are positive.
Finally, we show the total probability equals 1. For $j \in [0, q-1]$ such that $\rho(j) \neq 0$,

$$\sum_{i=0}^{q-1} \chi_j(i) = \sum_{i=0}^{q-1} \frac{\varphi_{j-i}(i)}{\sum\limits_{k=0}^{q-1} \varphi_k(j-k)} = \frac{\sum\limits_{i=0}^{q-1} \varphi_{j-i}(i)}{\sum\limits_{k=0}^{q-1} \varphi_k(j-k)} = \frac{\sum\limits_{k=0}^{q-1} \varphi_k(j-k)}{\sum\limits_{k=0}^{q-1} \varphi_k(j-k)} = 1.$$

B) Proof of the Theorem.

Let $\mathcal{A}$ be an adversary against the $\mathsf{LWE\text{-}OD}$ search problem with distributions $\varphi_0, \ldots, \varphi_{q-1}$. We need to build an adversary $\mathcal{B}$ against the qLWE search problem assuming that $\mathcal{B}$ has access to a $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ oracle with distributions $\chi_0, \ldots, \chi_{q-1}$ defined above and an unknown $\boldsymbol{k}$.

First, let $(\boldsymbol{a}, b)$ be the (non-$\perp$ with overwhelming probability) output of Algorithm 1 with input ρ and the given $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ sampler. Using Proposition 4, the output of Algorithm 1 is such that for any $j \in \mathbb{Z}_q$ and $\boldsymbol{l} \in \mathbb{Z}_q^n$,

$$\begin{aligned}
\Pr\left[\boldsymbol{a} = \boldsymbol{l}, b = j\right] &= \frac{1}{q^{n-1}} \cdot \chi_j(j - \langle \boldsymbol{l}, \boldsymbol{k} \rangle) \cdot \rho(j) \\
&= \frac{1}{q^{n-1}} \cdot \frac{\varphi_{\langle \boldsymbol{l}, \boldsymbol{k} \rangle}(j - \langle \boldsymbol{l}, \boldsymbol{k} \rangle)}{q \cdot \rho(j)} \cdot \rho(j) \\
&= \frac{1}{q^n} \cdot \varphi_{\langle \boldsymbol{l}, \boldsymbol{k} \rangle}(j - \langle \boldsymbol{l}, \boldsymbol{k} \rangle).
\end{aligned}$$

Hence as Proposition 4 shows that $\mathsf{LWE\text{-}OD}$ has the exact same distribution for the key $\boldsymbol{k}$, Algorithm 1 is indistinguishable from $\mathsf{LWE\text{-}OD}$ samples for the same key $\boldsymbol{k}$.
To conclude the proof, we explicitly construct the adversary $\mathcal{B}$ against the qLWE search problem as follows. Whenever $\mathcal{A}$ queries a sample from the $\mathsf{LWE\text{-}OD}$ distribution, $\mathcal{B}$ simulates this by running Algorithm 1 using its access to the $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ oracle (with distributions $\chi_0,\ldots,\chi_{q-1}$ and ρ as defined above). The resulting output $(\boldsymbol{a}, b)$ is returned to $\mathcal{A}$ as a $\mathsf{LWE\text{-}OD}^n_{\varphi_0,\ldots,\varphi_{q-1}}(\boldsymbol{k})$ sample. After a polynomial number of such queries, $\mathcal{A}$ outputs a candidate key $\boldsymbol{k} \in \mathbb{Z}_q^n$, which $\mathcal{B}$ also outputs.
By the previous analysis, the samples provided to $\mathcal{A}$ are distributed identically to true $\mathsf{LWE\text{-}OD}$ samples for the same secret key $\boldsymbol{k}$. Therefore, the success probability of $\mathcal{B}$ is directly related to that of $\mathcal{A}$, up to a polynomial loss due to the number of queries and the negligible probability of aborting. The overall time and memory complexity of $\mathcal{B}$ is polynomial in n, as it only performs a polynomial number of calls to the $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ oracle and polynomial-time computations. □

Remark 2. In the case of the theorem the distribution of $\boldsymbol{a}$ is uniform:

$$\Pr[\boldsymbol{a} = \boldsymbol{l}] = \sum_{j=0}^{q-1} \Pr[\boldsymbol{a} = \boldsymbol{l}, b = j] = \sum_{j=0}^{q-1} \varphi_{\langle \boldsymbol{l},\boldsymbol{k}\rangle}(j - \langle \boldsymbol{l},\boldsymbol{k}\rangle) \cdot \frac{1}{q^n} = \frac{1}{q^n}.$$

4.3 Step 2: a reduction from qLWE to LWE

In this section, we show that the $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ is at least as hard as $\mathsf{LWE}^n_{\psi}(\boldsymbol{k})$, under given conditions on the error distributions.

$$\mathsf{LWE} \xrightarrow{\text{Theorem 2}} \mathsf{qLWE}$$

The outline of the proof is quite similar to that in the previous section. We demonstrate that an adversary can convert $\mathsf{LWE}^n_{\psi}(\boldsymbol{k})$ samples into $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ samples without knowing the key $\boldsymbol{k}$ and within polynomial time. This conversion enables any attack on $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ to be applied to $\mathsf{LWE}^n_{\psi}(\boldsymbol{k})$, showing that the qLWE problem is at least as hard as the LWE problem.

We first give the sampler that uses the LWE sampler as a black-box, and give the conditions on the input distributions such that $\mathsf{LWE}^n_{\psi}(\boldsymbol{k})$ samples are indeed converted into $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ ones.
Intuitively, as the error of b_i', denoted e_i', follows the distribution ψ, one must find a distribution τ_i such that $e + e_i'$ follows the distribution χ_i. Adding two $\mathbb{Z}_q$-distribution χ_i and τ gives a convolution product. Indeed, let $X \leftarrow \chi_i, Y \leftarrow \tau$:

$$\forall k \in \mathbb{Z}_q, \Pr[X + Y = k] = \sum_{j \in \mathbb{Z}_q} \Pr[X + Y = k \mid X = j] \cdot \Pr[X = j]$$

Algorithm 2: AltSample-q-LWE

Input: $\mathbb{Z}_q$-distributions $\tau_0, \ldots, \tau_{q-1}$, a $\mathsf{LWE}^n_\psi(\boldsymbol{k})$ sampler
Output: A sample $(\boldsymbol{a_i}, b_i)_{0\le i<q} \in \left(\mathbb{Z}_q^n \times \mathbb{Z}_q\right)^q$.

```
for i in range(q) do
    (a_i, b'_i) ← SampleLWE^n_ψ(k);
    e ← τ_i
    b_i := b'_i + e
end
return (a_i, b_i)_{0≤i<q}
```

$$= \sum_{j\in\mathbb{Z}_q} \tau(k-j)\cdot\chi_i(j)$$
$$= (\chi_i * \ \tau)\,(k),$$

where $*$ denotes the convolution product.
In other words, to obtain a qLWE sampler, one must find $\tau_0, \cdots, \tau_{q-1}$ such that for all $i \in [0, q-1]$,

$$\tau_i * \psi = \chi_i.$$

For an arbitrary ψ, it remains into solving the *deconvolution problem* for discrete distributions. One standard technique for solving this problem is the discrete Fourier analysis, as this analysis transforms convolutions into products. While it is always possible to find a discrete function τ_i that verifies the convolution, this function is not always a distribution. Not only it might not sum to 1 but it is actually not always in $\mathbb{R}$ as the Fourier transform can have complex values. Therefore, in the next section, we provide constraints on ψ and χ for the reduction to work. If the constraints are not verified, we could not prove the final reduction.

Definition 7 (Walsh transform). *Let ω be a q-th root of unity. For any $\mathbb{Z}_q$ distribution $\mathcal{D}$, we define its Walsh transform $\widehat{\mathcal{D}}$ as*

$$\begin{aligned}\widehat{\mathcal{D}} :&\mathbb{Z}_q \to \mathbb{C}\\ i &\mapsto \sum_{k=0}^{q-1} \omega^{ik}\mathcal{D}(k)\end{aligned}$$

Property 1 (Properties of the Walsh transform). The Walsh transform has the following properties.

1. Let $\mathcal{D}_0$ and $\mathcal{D}_1$ be two $\mathbb{Z}_q$-distributions. For any $i \in \mathbb{Z}_q$, $\widehat{\mathcal{D}_0 * \mathcal{D}_1}(i) = \widehat{\mathcal{D}_0}(i)\cdot\widehat{\mathcal{D}_1}(i)$.
2. Let $\mathcal{D}$ be an $\mathbb{Z}_q$-distribution. It is possible to invert the Walsh transform with $\mathcal{D}(j) = \frac{1}{q}\sum_{k\in\mathbb{Z}_q} \widehat{\mathcal{D}}(j)\omega^{-jk} \in \mathbb{C}$ for $j \in \mathbb{Z}_q$.

The proof of this property is well-known and can be found in Appendix, Section A.

Proposition 5 (Algorithm 2 output characterization). *Let $\boldsymbol{k} \in (\mathbb{Z}_q^n)^*$ be a secret vector. We suppose an access to Sample$\mathsf{LWE}_\psi^n(\boldsymbol{k})$. Let $\chi_0, \ldots, \chi_{q-1}$ be a collection of q distributions. Let ψ be a $\mathbb{Z}_q$-distribution such that $\bigcup_{i \in \mathbb{Z}_q} Supp(\widehat{\chi}_i) \subseteq Supp(\widehat{\psi})$. Let $\tau_0, \ldots, \tau_{q-1}$ be q distributions on $\mathbb{Z}_q$ such that:*

$$\forall i, j \in \mathbb{Z}_q, \begin{cases} \tau_i(j) = \frac{1}{q} \sum_{k \in Supp(\widehat{\chi}_i)} \frac{\widehat{\chi_i}(k)}{\widehat{\psi}(k)} \omega^{-jk} \\ \tau_i(j) \in \mathbb{R}_{\geq 0} \end{cases}$$

where $\omega = exp\left(\frac{2i\pi}{q}\right)$ is an (arbitrary) q-th root of unity and $\widehat{\chi}_i$ [resp. $\widehat{\psi}$] is the Walsh transform of χ_i [resp. ψ]. Then, the output of Algorithm 2 with input $\tau_0, \ldots, \tau_{q-1}$ follows a $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ distribution.

In a nutshell, the function τ_i is defined as the solution to the deconvolution problem $\tau_i * \psi = \chi_i$ using the Walsh transform. The condition on the support ensures that there is no division by zero when defining τ_i. Finally, we need to ensure that τ_i is indeed a distribution, which is done by the second condition.

Proof. **A) The distributions are well defined.**
By hypothesis, since $\bigcup_{i \in \mathbb{Z}_q} \text{Supp}(\widehat{\chi}_i) \subseteq \text{Supp}(\widehat{\psi})$, $\widehat{\psi}(k) \neq 0$ for all $k \in \text{Supp}(\widehat{\chi}_i)$ for any $i \in \mathbb{Z}_q$, hence, $\tau_0, \cdots, \tau_{q-1}$ are correct functions. Let us verify that they are distributions over $\mathbb{Z}_q$ (i.e. $\tau_i(j) \in [0,1]$ and $\sum_k \tau_i(k) = 1$ for $i, j \in \mathbb{Z}_q$).
Let us first check that the probabilities sum at 1. We have:

$$\sum_{k=0}^{q-1} \tau_i(k) = \widehat{\tau}_i(0).$$

By applying the Walsh transform to τ_i, we obtain for all $i \in \mathbb{Z}_q$:

$$\widehat{\tau}_i(0) = \frac{\widehat{\chi}_i(0)}{\widehat{\psi}(0)} = 1$$

as χ_i and ψ are distributions.

In addition, as by hypothesis, $\tau_i(k) \geq 0 \ \forall k \in \mathbb{Z}_q$ and $\sum_{k=0}^{q-1} \tau_i(k) = 1$, we can conclude that $\tau_i(k) \leq 1 \ \forall k \in \mathbb{Z}_q$.

B) Proof of the proposition.
Algorithm 2 does not contain any rejection, hence one can see that the output follows a $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ distribution if and only if

$$\forall i \in \mathbb{Z}_q, \ \tau_i * \psi = \chi_i.$$

Let us check that $\tau_0, \ldots, \tau_{q-1}$ as defined in the proposition validate this equation. By applying the Walsh transform to τ_i, we obtain for all $i \in \mathbb{Z}_q$ and $j \in \text{Supp}(\widehat{\chi}_i)$:

$$\widehat{\tau}_i(j) = \sum_{k' \in [0,q-1]} \frac{1}{q} \sum_{k \in \text{Supp}(\widehat{\chi}_i)} \frac{\widehat{\chi}_i(k)}{\widehat{\psi}(k)} \omega^{-k'k} \omega^{jk'}$$

$$= \frac{1}{q} \sum_{k \in \mathrm{Supp}(\widehat{\chi}_i)} \frac{\widehat{\chi}_i(k)}{\widehat{\psi}(k)} \sum_{k' \in [0,q-1]} \omega^{(j-k)k'}$$

$$= \frac{1}{q} \sum_{k \in \mathrm{Supp}(\widehat{\chi}_i)} \frac{\widehat{\chi}_i(k)}{\widehat{\psi}(k)} q \cdot \delta_{j,k}$$

$$= \frac{\widehat{\chi}_i(j)}{\widehat{\psi}(j)}.$$

Hence, for all $i, j \in \mathbb{Z}_q$,

$$\widehat{\tau}_i \cdot \widehat{\psi} = \widehat{\chi}_i.$$

Using item 1 of Property 1,

$$\widehat{\tau}_i \cdot \widehat{\psi} = \widehat{\chi}_i \iff \tau_i * \psi = \chi_i,$$

which concludes the proof. □

We can now use this lemma to give the following reduction:

Theorem 2 (LWE reduction to qLWE). *Let $\psi, \chi_0, \ldots, \chi_{q-1}$ be $\mathbb{Z}_q$-distributions with q polynomial in n. We introduce the following conditions.*

1. $\bigcup_{i \in \mathbb{Z}_q} \mathit{Supp}(\widehat{\chi}_i) \subseteq \mathit{Supp}(\widehat{\psi})$,
2. $\forall i, j \in \mathbb{Z}_q$,

$$\frac{1}{q} \sum_{\substack{k \in \mathbb{Z}_q \\ \widehat{\psi}(k) \neq 0}} \frac{\widehat{\chi}_i(k)}{\widehat{\psi}(k)} \omega^{-jk} \in \mathbb{R}_{\geq 0}.$$

For any adversary $\mathcal{A}$ against the qLWE search problem, there exists an adversary $\mathcal{B}$ against the LWE search problem such that:

$$\boxed{\Pr[\boldsymbol{k} \leftarrow (\mathbb{Z}_q^n)^* : \mathcal{A}^{\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})}(1^n) = \boldsymbol{k}] \leq q \cdot \Pr[\boldsymbol{k} \leftarrow (\mathbb{Z}_q^n)^* : \mathcal{B}^{\mathsf{LWE}^n_{\psi}(\boldsymbol{k})}(1^n) = \boldsymbol{k}],}$$

and $\mathcal{A}$ and $\mathcal{B}$ both have polynomial memory and time complexity.

Proof. Let $\mathcal{A}$ be an adversary against the qLWE search problem with distributions $\chi_0, \ldots, \chi_{q-1}$. We need to build an adversary $\mathcal{B}$ against the LWE search problem assuming that $\mathcal{B}$ can query an $\mathsf{LWE}^n_{\psi}(\boldsymbol{k})$ oracle with the distribution ψ defined above and an unknown $\boldsymbol{k}$.

First, we define the distributions $\tau_0, \ldots, \tau_{q-1}$ as follows:

$$\tau_i(x) = \sum_{k \in \mathbb{Z}_q} \frac{\widehat{\chi}_i(k)}{\widehat{\psi}(k)} \cdot \omega^{-kx} \text{ for } x \in \mathbb{Z}_q.$$

The first condition of the theorem ensures that $\tau_0, \ldots, \tau_{q-1}$ are well defined functions Indeed, $\bigcup_{i \in \mathbb{Z}_q} \mathrm{Supp}(\widehat{\chi}_i) \subseteq \mathrm{Supp}(\widehat{\psi})$ prevents any potential division by zero. The second condition ensures that they are indeed distributions over $\mathbb{Z}_q$.
We construct the adversary $\mathcal{B}$ as follows. Given access to an $\mathsf{LWE}^n_{\psi}(\boldsymbol{k})$ oracle with secret $\boldsymbol{k}$ and error distribution ψ, $\mathcal{B}$ simulates the $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$

oracle for $\mathcal{A}$ by using Algorithm 2 with the distributions $\tau_0, \ldots, \tau_{q-1}$. Specifically, whenever $\mathcal{A}$ queries a $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ sample, $\mathcal{B}$ queries its $\mathsf{LWE}^n_{\psi}(\boldsymbol{k})$ oracle q times to obtain the q independent samples, adds an independent error drawn from τ_i to each b'_i, and returns the resulting $(\boldsymbol{a}_i, b_i)$ to $\mathcal{A}$.
When $\mathcal{A}$ outputs a guess $\boldsymbol{k}^*$ for the secret, $\mathcal{B}$ outputs the same guess.
By construction and by Proposition 5, the simulated $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ samples are distributed exactly as required under the theorem's conditions. Thus, the success probability of $\mathcal{A}$ in recovering $\boldsymbol{k}$ is preserved up to a factor q due to the q-to-1 correspondence between $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ and $\mathsf{LWE}^n_{\psi}(\boldsymbol{k})$ samples. The time and memory complexity of $\mathcal{B}$ is polynomial, as it only requires polynomially many calls to the $\mathsf{LWE}^n_{\psi}(\boldsymbol{k})$ oracle and polynomial-time post-processing.
This completes the reduction. $\square$

Remark 3. Item 1 of Theorem 2 is automatically validated as long as $\widehat{\psi}(k) \neq 0$ for all $k \in \mathbb{Z}_q$.

4.4 Final reduction and remarks

Finally, we can close up the reduction by combining Theorem 2 and Theorem 1:

Theorem 3 (LWE to LWE-OD). *Let $\varphi_0, \cdots, \varphi_{q-1}$ be $\mathbb{Z}_q$-distributions with q polynomial in n. Let ψ be another $\mathbb{Z}_q$-distribution validating the following conditions:*

1. $\bigcup_{i \in \mathbb{Z}_q} S_i \subseteq \mathit{Supp}(\widehat{\psi})$ *, with* $S_i := \{k \text{ such that } \widehat{\varphi_{i-\cdot}(\cdot)}(k) \neq 0\}$
2. $\forall i \in \mathbb{Z}_q$ *such that* $\sum_{u=0}^{q-1} \varphi_u(i-u) \neq 0$, $\forall j \in \mathbb{Z}_q$
$$\sum_{\ell=0}^{q-1} \varphi_{i-\ell}(\ell) \sum_{\substack{k \in \mathbb{Z}_q \\ \widehat{\psi}(k) \neq 0}} \frac{\omega^{k\cdot(\ell-j)}}{\widehat{\psi}(k)} \in \mathbb{R}_{\geq 0}.$$

For any adversary $\mathcal{A}$ against the LWE-OD *search problem, there exists an adversary $\mathcal{B}$ against the* LWE *search problem such that:*

$$\boxed{\begin{aligned}&\Pr[\boldsymbol{k} \leftarrow (\mathbb{Z}_q^n)^* : \mathcal{A}^{\mathsf{LWE\text{-}OD}^n_{\varphi_0,\ldots,\varphi_{q-1}}(\boldsymbol{k})}(1^n) = \boldsymbol{k}] \\ &\quad \leq m \cdot q \cdot \Pr[\boldsymbol{k} \leftarrow (\mathbb{Z}_q^n)^* : \mathcal{B}^{\mathsf{LWE}^n_{\psi}(\boldsymbol{k})}(1^n) = \boldsymbol{k}],\end{aligned}}$$

where m is a polynomial large enough to grow at least proportionally to $q \cdot \log(n)$ and $\mathcal{A}$ and $\mathcal{B}$ both have the polynomial memory and time complexity.

Proof. The proof follows directly by composing the reductions established in Theorems 1 and 2.
First, by Theorem 1, any adversary $\mathcal{A}$ against the LWE-OD problem can be transformed into an adversary $\mathcal{A}'$ against the qLWE problem, with ρ and the χ_js as stated in Theorem 1.

Second, Theorem 2, any adversary $\mathcal{A}'$ against the qLWE problem can be transformed into an adversary $\mathcal{B}$ against the LWE problem, under the conditions stated in the theorem.
By composing these two reductions, we obtain that any adversary $\mathcal{A}$ against the $\mathsf{LWE\text{-}OD}$ problem can be transformed into an adversary $\mathcal{B}$ against the LWE problem, with the overall loss in advantage being the product of the losses in each reduction (i.e., a polynomial factor $m \cdot q$). Both reductions preserve polynomial time and memory complexity.
Therefore, we combine the conditions of Theorem 1 with the definition of ρ and χ_j as stated in Theorem 2 to obtain the following condition on $\psi, \varphi_0, \cdots, \varphi_{q-1}$: $\forall i \in \mathbb{Z}_q$ such that $\sum_{u=0}^{q-1} \varphi_u(i-u) \neq 0$, $\forall j \in \mathbb{Z}_q$

$$\frac{1}{q \cdot \sum_{u=0}^{q-1} \varphi_u(i-u)} \cdot \sum_{\substack{k \in \mathbb{Z}_q \\ \widehat{\psi}(k) \neq 0}} \sum_{\ell=0}^{q-1} \frac{\varphi_{i-\ell}(\ell)}{\widehat{\psi}(k)} \omega^{k \cdot (\ell - j)} \in \mathbb{R}_{\geq 0}.$$

One can simplify this condition by removing $\frac{1}{q \cdot \sum_{u=0}^{q-1} \varphi_u(i-u)} \in \mathbb{R}_{\geq 0}$ and changing the order of the sums. $\square$

Theorem 3 provides a condition for efficiently verifying whether a given $\mathsf{LWE\text{-}OD}$ instance can be reduced from another given LWE instance. It is novel and allows extending our knowledge on the LWE in the case of extra knowledge. More specifically, as shown in the next Section, it can be used to recover specific reductions like LWR to LWE (see Section 5.1) and, it can also be applied for inexact computing and with specific distributions (see Section 5.2). However, one should note that Theorem 3 has certain limitations:

- For any given LWE instance, i.e., a known ψ, it is not straightforward to find non-trivial family of $\mathsf{LWE\text{-}OD}$ instances that are at least as hard as the LWE instance. This case is not the most important one as one would typically want to reduce more complex distributions from simpler ones like LWE, but it highlights that the reduction is not easy to invert.
- Similarly, for any given $\mathsf{LWE\text{-}OD}$ instance, several ψ can validate the condition and it is not immediately clear how to find the best LWE distribution (i.e. find the ψ that minimizes the LWE advantage) that it reduces from.

Some trivial reductions of Theorem 3 are given in Appendix, Section B.

Remark 4. Any $\mathsf{LWE\text{-}OD}$ instance can be reduced from the LWE distribution with any distribution ψ such that for any $t \in \mathbb{Z}_q$, $\widehat{\psi}(k) \in \mathbb{R}$ and:

$$F(t) := \sum_{\substack{k \in \mathbb{Z}_q \\ \widehat{\psi}(k) \neq 0}} \frac{\cos\left(\frac{2\pi k t}{q}\right)}{\widehat{\psi}(k)} \geq 0.$$

Indeed, in that case, one can verify that the conditions of Theorem 3 are validated.

However, as expected (because of the absence on condition on φ), it is not possible to find such functions. Indeed, $F(t) \geq 0$ implies that the positive cosines of the sum should get high weight. Hence, $k \mapsto \frac{1}{\widehat{\psi}(k)}$ should be concentrated around 0. So, this implies that $k \mapsto \widehat{\psi}(k)$ should be concentrated around $q/2$. And, applying the Walsh transform the high weight will this time be on the negative cosines, hence not ending into a distribution.

5 Applications

5.1 The LWR particular case

As shown in Proposition 1, the LWR distribution can be seen as a particular case of LWE-OD distribution. Plugging this distribution into Theorem 3, we can make a reduction from LWE to LWR. By doing so, we get an analog of the result of [BGM+16, Section 4] for the search variant. While it is not totally surprising as solving the search variant implies solving the decision variant , our result includes more generality as it captures not only $p|q$ but also the general case using a mild conjecture. First, we present the particular case of $p|q$ that is an application of Theorem 3.

Corollary 1 (LWE $\implies$ LWR when $p|q$). *Let n be an integer. Let $p < q$ be integers such that $p|q$. The $\mathsf{LWR}_n^{p,q}$ search problem is at least as hard as the LWE_n^{ψ} one with $\psi = \mathcal{U}\left(\left[-\frac{q}{p}+1, 0\right]\right)$.*

Proof. Let us define $L := \frac{q}{p} \in \mathbb{Z}$ for simplifying the computations. Let $\psi = \mathcal{U}([-L+1, 0])$. Firstly, let us compute $\widehat{\psi}$.

$$\begin{aligned}
\widehat{\psi}(k) &= \sum_{\ell=0}^{q-1} \omega^{k\ell}\psi(\ell) \\
&= \sum_{\ell=-L+1}^{0} \frac{1}{L}\omega^{k\ell} \\
&= \begin{cases} 1 \text{ if } k = 0, \\ \frac{\omega^{k(-L+1)}}{L} \frac{1-\omega^{kL}}{1-\omega^k} \text{ else.} \end{cases}
\end{aligned}$$

Now, as stated in Proposition 1, the distribution φ that corresponds to LWR is as follows. For each $u \in [0, q-1]$:

$$\varphi_u(x) = \begin{cases} 1 & x = \mathrm{round}_p(u) - u \bmod q, \\ 0 & \text{otherwise.} \end{cases}$$

Next, let us fix $i \in [0, q-1]$, we have

$$\begin{aligned}
\varphi_{i-\ell_i \bmod q}(\ell_i) = 1 &\iff \left\lfloor \frac{p}{q}(i - \ell_i \bmod q) \right\rfloor - i + \ell_i = \ell_i \bmod q \\
&\iff \left\lfloor \frac{p}{q}(i - \ell_i \bmod q) \right\rfloor = i
\end{aligned}$$

We note that it implies that there are no solutions if $i \geq p$. For $i \in [0, p-1]$, we have

$$\begin{aligned}
\varphi_{i-\ell_i \bmod q}(\ell_i) = 1 \iff & \; i \leq \frac{p}{q}\,(i - \ell_i \bmod q) < i+1 \\
\iff & \; i \leq \frac{p}{q}\,(i - \ell_i + Cq) < i+1 \text{ for some C} \in \mathbb{Z}, \\
\iff & \; i\,(1-L) - L + Cq < \ell_i \leq i\,(1-L) + Cq.
\end{aligned}$$

The value $C = 1$ is the only possible solution for $\ell_i \in [0, q]$. Indeed, recall that $i \leq p-1$ and $L \geq 1$ (because $p \neq q$), we get $i\,(1-L) + q \leq q-1$ and $i\,(1-L) - L + q = i\,(1-L) - L + Lp = i\,(1-L) + L(p-1) \geq (p-1)\,(1-L) + L(p-1) \geq p-1 \geq 0$.
In other words,

$$\varphi_{i-\ell_i}(\ell_i) = 1 \iff i \leq p \text{ and } \ell_i \in \left]i\,(1-L) - L + q, i\,(1-L) + q\right].$$

We note that the interval contains exactly L integer values.
We now apply Theorem 3. The first condition is automatically validated as $\widehat{\psi}(k) \neq 0$, hence $\mathrm{Supp}(\widehat{\psi}) = [0, q-1]$. Let us compute the value for the second condition. For $i \in [0, p-1]$ and $j \in [0, q-1]$,

$$\begin{aligned}
& \sum_{\ell=0}^{q-1} \varphi_{i-\ell \bmod q}(\ell) \sum_{\substack{k=0 \\ \widehat{\psi}(k) \neq 0}}^{q-1} \frac{\omega^{k\cdot(\ell-j)}}{\widehat{\psi}(k)} \\
&= \sum_{\ell=0}^{q-1} \sum_{k=0}^{q-1} \varphi_{i-\ell \bmod q}(\ell) \frac{\omega^{k\cdot(\ell-j)}}{\widehat{\psi}(k)} \\
&= \sum_{\ell=0}^{q-1} \varphi_{i-\ell \bmod q}(\ell) \frac{1}{\widehat{\psi}(0)} + \sum_{k=1}^{q-1} \sum_{\ell=0}^{q-1} \varphi_{i-\ell \bmod q}(\ell) \frac{\omega^{k\cdot(\ell-j)}}{\widehat{\psi}(k)}
\end{aligned}$$

We next apply the expression of $\widehat{\psi}(k)$ computed at the beginning of the proof.

$$\begin{aligned}
& \sum_{\ell=0}^{q-1} \varphi_{i-\ell \bmod q}(\ell) \sum_{\substack{k=0 \\ \widehat{\psi}(k) \neq 0}}^{q-1} \frac{\omega^{k\cdot(\ell-j)}}{\widehat{\psi}(k)} \\
&= \sum_{\ell=0}^{q-1} \varphi_{i-\ell \bmod q}(\ell) \; + \; L \sum_{k=1}^{q-1} \frac{1-\omega^k}{1-\omega^{kL}} \sum_{\ell} \varphi_{i-\ell \bmod q}(\ell)\, \omega^{k(\ell-j+L-1)} \\
&= L + L \sum_{k=1}^{q-1} \frac{1-\omega^k}{1-\omega^{kL}} \sum_{\ell=i(1-L)-L+q}^{i(1-L)+q} \omega^{k(\ell-j+L-1)} \\
&= L + L \sum_{k=1}^{q-1} \frac{1-\omega^k}{1-\omega^{kL}} \frac{1-\omega^{kL}}{1-\omega^k} \omega^{k(-j-1+i(1-L))} \\
&= L + L \sum_{k=1}^{q-1} \omega^{k(-j-1+i(1-L))} \in \{0, L\}.
\end{aligned}$$

The second condition of Theorem 3 is successfully validated. $\square$

Link to [BGM+16]. Our result is similar to the one in [BGM+16] for the search variant. Our result in a little less greedy as their equivalent of Algorithm 2 contains rejections. Explicitly, their sampling method corresponds to Algorithm 3 using our notations.

Algorithm 3: LWR sampler from [BGM+16]

Input: An $\mathsf{LWE}^n_\psi(\boldsymbol{k})$ sampler
Output: An $\mathsf{LWR}^n_{p,q}(\boldsymbol{k})$ sample $(\boldsymbol{a}, b) \in \mathbb{Z}_q^n \times \mathbb{Z}_p$.
1 **do**
2 $\quad (\boldsymbol{a}, b) \leftarrow$ Sample-$\mathsf{LWE}^n_\psi(\boldsymbol{k})$;
3 **while** $\frac{q}{p} \nmid b$;
4 **return** $(\boldsymbol{a}, b \times (p/q))$

More general result In the following, we provide a more general case based on the following conjecture.

Conjecture 1. Let n and $a < 0$ be integer parameters. The LWE^ψ_n search problem with $\psi = \mathcal{U}([a, 0])$ is at least as hard as LWE^ψ_n search problem with $\psi = \mathcal{U}([a+1, 0])$.

This conjecture is validated in practice as the common belief is the more error, the more difficult the problem. More precisely, the current best cryptanalysis is based on lattice reduction techniques (e.g. [ACD+18,DDGR20]). And, the difficulty of lattice reduction depends only on the standard deviation of the noise and not the exact shape of the error distribution. Nevertheless, there is no known black-box proof of this reduction. Indeed, in all generality, a reduction should take into account an attacker for LWE search problem with $\psi = \mathcal{U}([a+1, 0])$ who would potentially succeed only if the error is equal to $a+1$. An argument using the statistical distance (or other statistical measures like smooth Rényi divergence) between both distribution is not strong enough as the integer a is polynomial in the security parameter, the advantage loss is too high for a correct reduction.

Theorem 4 ($\mathsf{LWE} \implies \mathsf{LWR}$ (general case)). *Assuming Conjecture 1, let $n \in \mathbb{N}$ and $p \leq q$ be integers. Let $L := \left\lfloor \frac{q}{p} \right\rfloor$. The $\mathsf{LWR}^{p,q}_n$ search problem is at least as hard as the LWE^ψ_n search problem with $\psi = \mathcal{U}(]-L+1, 0])$.*

Proof. **A) Application of Theorem 1.**
As stated in Proposition 1 and in the proof of Corollary 1, the distributions φ verify:

$$\forall i, j \in \mathbb{Z}_q\ \varphi_{j-i \bmod q}(i) = 1 \iff j \leq p-1 \text{ and } j\left(1-\frac{q}{p}\right)-\frac{q}{p}+q < i \leq j\left(1-\frac{q}{p}\right)+q.$$

We apply Theorem 1. When $j \geq p$, $\rho(j) = 0$, hence, χ_j can be any distribution. By simplicity, it is chosen as the same as when $j \leq p-1$. We obtain for $j \leq q-1$,

$$\chi_j = \mathcal{U}\left(\left[\ \left\lceil j\left(1-\frac{q}{p}\right) - \frac{q}{p} + q\right\rceil, \left\lfloor j\left(1-\frac{q}{p}\right) + q\right\rfloor\ \right]\right).$$

We write $j\left(1-\frac{q}{p}\right) = j - \left\lfloor \frac{jq}{p} \right\rfloor + \epsilon_0$ and $\frac{q}{p} = L + \epsilon_1$, with $0 \leq \epsilon_0, \epsilon_1 < 1$. If $p|q$, $\epsilon_0 = \epsilon_1 = 0$. Else if $p|jq$, $\epsilon_0 = 0 \neq \epsilon_1$. The interval is of size:

$$\left\lfloor j\left(1-\frac{q}{p}\right) + q\right\rfloor - \left\lceil j\left(1-\frac{q}{p}\right) - \frac{q}{p} + q\right\rceil = L - \lceil \epsilon_0 - \epsilon_1 \rceil = \begin{cases} L \text{ if } \epsilon_0 \geq \epsilon_1, \\ L-1 \text{ else.} \end{cases}$$

Hence, the interval is of size either L or $L-1$, depending on the sign of $\epsilon_0 - \epsilon_1 = \left\lfloor \frac{jq}{p} \right\rfloor + L - \frac{jq}{p} - \frac{q}{p} \in \mathbb{R}$. Note that if $q/p \in \mathbb{Z}$, it is always of size L and we fall back to Corollary 1.

B) Application of Conjecture 1.
Applying the conjecture, it means that there exists a polynomial reduction algorithm that, given a polynomial access to an $\mathsf{LWE}^n_{\mathcal{U}(]-L+1,0])}(\boldsymbol{k})$ sampler, can build a instance that follows the distribution $\mathsf{LWE}^n_{\mathcal{U}([-L+1,0])}(\boldsymbol{k})$. Let us denote it $\mathcal{R}^{\mathsf{LWE}^n_{\mathcal{U}(]-L+1,0])}(\boldsymbol{k})}$.

C) Application of a variant of Theorem 2.
Now given an access to an $\mathsf{LWE}^n_{\mathcal{U}(]-L+1,0])}(\boldsymbol{k})$ sampler, we need to build a sample that follows $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ with χ_j defined above in **A)**. For that, we proceed following Algorithm 4. First, the conjecture is used to transform some samples to obtain the correct support. For each sample $j \in \mathbb{Z}_q$, one can separate two cases: in the cases when $\left\lfloor \frac{jq}{p} \right\rfloor + L - \frac{jq}{p} - \frac{q}{p} \geq 0$, the sample is drawn from $\mathcal{R}^{\mathsf{LWE}^n_{\mathcal{U}(]-L+1,0])}(\boldsymbol{k})}$, therefore following $\mathsf{LWE}^n_{\mathcal{U}([-L+1,0])}(\boldsymbol{k})$ and in the other case, the sample is simply drawn from $\mathsf{LWE}^n_{\mathcal{U}(]-L+1,0])}(\boldsymbol{k})$. Next, a shift is applied to end up with the final χ_j interval. The number of calls to the oracle $\mathsf{LWE}^n_{\mathcal{U}(]-L+1,0])}(\boldsymbol{k})$ sampler is polynomial in n which concludes the proof.
By construction, Algorithm 4 correctly samples $\mathsf{qLWE}^n_{\chi_0,\ldots,\chi_{q-1}}(\boldsymbol{k})$ with χ_j defined above in **A)**. This algorithm calls $\mathsf{LWE}^n_{\mathcal{U}(]-L+1,0])}(\boldsymbol{k})$ a polynomial number of times which concludes the proof.

□

5.2 Application to inexact computing

First, we use Theorem 3 to verify [BHK+21, Corollary 1], which is the dimension 2 case. In this case, every error distribution is a Bernoulli distribution.

Corollary 2 ([BHK+21]). *Let $q = 2$, for $n \in \mathbb{N}$, $\varepsilon, \Delta \neq 0$ error parameters such that $0 \leq \varepsilon - \Delta \leq \varepsilon + \Delta \leq \frac{1}{2}$, we define $\varphi_0 = \mathsf{Ber}_{\varepsilon-\Delta}$ and $\varphi_1 = \mathsf{Ber}_{\varepsilon+\Delta}$. $\mathsf{LPN\text{-}OD}^n_{\varphi_0,\varphi_1}$ is at least as hard as LPN^n_{ψ} for $\psi = \mathsf{Ber}_{\frac{\varepsilon-\Delta}{1-2\Delta}}$.*

Algorithm 4: Sample-q-LWE-double-case

Input: an $\mathsf{LWE}^n_{\mathcal{U}(]-L+1,0])}(\boldsymbol{k})$ sampler

Output: A sample $(\boldsymbol{a_i}, b_i)_{0\leq i<q} \in \left(\mathbb{Z}_q^n \times \mathbb{Z}_q\right)^q$.

for $j \leftarrow 0$ **to** $q-1$ **do**
  if $\left\lfloor \frac{jq}{p} \right\rfloor + L - \frac{jq}{p} - \frac{q}{p} \geq 0$ **then**
    $(\boldsymbol{a_j}, b_j) \leftarrow \mathcal{R}^{\mathsf{LWE}^n_{\mathcal{U}(]-L+1,0])}(\boldsymbol{k})}$ (Conjecture 1)
  else
    $(\boldsymbol{a_j}, b_j) \leftarrow \mathsf{LWE}^n_{\mathcal{U}(]-L+1,0])}(\boldsymbol{k})$
  end
  $b_j \leftarrow b_j + \left\lfloor j\left(1-\frac{q}{p}\right) + q \right\rfloor \bmod q$
end
return $(\boldsymbol{a_j}, b_j)_{0\leq j<q}$

Proof. First, let us remark that $\mathsf{LPN\text{-}OD}^n_{\varphi_0,\varphi_1}$ and LPN^n_ψ are the particular case of $q=2$ of $\mathsf{LWE\text{-}OD}^n_{\varphi_0,\varphi_1}$ and LWE^n_ψ. Hence, Theorem 3 applies.
Firstly, let us compute $\widehat{\psi}$. In the case of $q=2$, ω is simply -1.

$$\begin{aligned}
\widehat{\psi}(0) &= \psi(0) + \psi(1) = 1,\\
\widehat{\psi}(1) &= \psi(0) - \psi(1)\\
&= 1 - \frac{\varepsilon - \Delta}{1-2\Delta} - \frac{(\varepsilon-\Delta)}{1-2\Delta}\\
&= \frac{1-2\varepsilon}{1-2\Delta}.
\end{aligned}$$

The conditions on ε and Δ imply $\varepsilon \neq 1/2$. Therefore, we have $\widehat{\psi}(0), \widehat{\psi}(1) \neq 0$. The first condition is validated.
Now, let $i,j \in \{0,1\}$, let us compute the second condition.

$$\begin{aligned}
\sum_{\ell=0}^{1} \varphi_{i-\ell}(\ell) \sum_{k=0}^{1} \frac{(-1)^{k\cdot(\ell-j)}}{\widehat{\psi}(k)} &= \varphi_i(0)\left(1 + \frac{(-1)^{-j}}{\widehat{\psi}(1)}\right) + \varphi_{i-1}(1)\left(1 + \frac{(-1)^{1-j}}{\widehat{\psi}(1)}\right)\\
&= \varphi_{i-1}(1) + \varphi_i(0) + \frac{\varphi_{i-1}(1)(-1)^{1-j} + \varphi_i(0)(-1)^{-j}}{\widehat{\psi}(1)}\\
&= \varphi_{i-1}(1) + \varphi_i(0) + (1-2\Delta)\frac{\varphi_{i-1}(1)(-1)^{1-j} + \varphi_i(0)(-1)^{-j}}{1-2\varepsilon}\\
&= \begin{cases} 2 \text{ if } i,j=0,\\ 4\Delta \text{ if } i=0, j=1,\\ 2-4\Delta \text{ if } i=1, j=0,\\ 0 \text{ if } i=1, j=1. \end{cases}
\end{aligned}$$

All these values are positive, which verifies the theorem's conditions and proves the reduction. $\square$

Despite this positive result with a binary modulus ($q = 2$), the theorem proves challenging to apply directly to the LWPE setting. Fundamentally, physical errors are difficult to model exactly. Hardware observations suggest defining LWPE as an $\mathsf{LWE\text{-}OD}$ variant characterized by "close" error distributions. Informally, this means $\mathsf{LWPE}^n_\eta(\boldsymbol{k}) \equiv \mathsf{LWE\text{-}OD}^n_{\varphi_0,\ldots,\varphi_{q-1}}(\boldsymbol{k})$ where the distributions satisfy $\forall i, j, \Delta(\varphi_i, \varphi_j) \leq \eta$ for some statistical distance Δ. However, working with such an inequality condition on the error distributions makes the requirements of Theorem 3 computationally intractable. Our simulator-based approach, particularly the critical step that reduces LWE to qLWE, relies instead on a strict equality of distributions. While this reduction is feasible for $q = 2$ because all square roots of unity are real numbers, it constitutes a significant bottleneck when attempting to generalize the analysis to higher moduli ($q > 2$).

6 Conclusion and open problems

In this work, we introduce and provide an initial analysis of a generalized learning problem called Learning with Errors with Output Dependencies ($\mathsf{LWE\text{-}OD}$). This problem is defined in such a way that it encompasses traditional problems like LWE and LWR, but also some interesting physical learning problems such as $\mathsf{LPN\text{-}OD}$. These unconventional problems have useful applications in inexact computing and leveled implementation analysis, making the question of their hardness a promising topic.
There are several directions for future research stemming from this work. The main issue of this work is the fact that for a given $\mathsf{LWE\text{-}OD}$ instance, it is not immediately clear how to find the tightest reduction, i.e., the LWE distribution that minimizes the LWE advantage. Exploring this question further is ambitious but could lead to a better understanding of the relationships between different learning problems and their respective hardness. As another more practical direction, it would be interesting to investigate whether our framework can be used to establish a reduction from LWE to LWPR, which would have significant implications for the security of cryptographic schemes based on LWPR.

A Property 1 proof

Proof. 1. We show that $\widehat{\mathcal{D}_0 * \mathcal{D}_1}(i) = \widehat{\mathcal{D}_0}(i) \cdot \widehat{\mathcal{D}_1}(i)$. Let $i \in \mathbb{Z}_q$:

$$\begin{aligned}
\widehat{\mathcal{D}_0 * \mathcal{D}_1}(i) &= \sum_{k \in \mathbb{Z}_q} \omega^{ik} \left(\sum_{j \in \mathbb{Z}_q} \mathcal{D}_0(j) \mathcal{D}_1(k-j) \right) \\
&= \sum_{j \in \mathbb{Z}_q} \sum_{k \in \mathbb{Z}_q} \omega^{ik} \mathcal{D}_0(j) \mathcal{D}_1(k-j) \\
&= \sum_{j \in \mathbb{Z}_q} \sum_{k' \in \mathbb{Z}_q} \omega^{i(k'+j)} \mathcal{D}_0(j) \mathcal{D}_1(k')
\end{aligned}$$

$$= \sum_{k' \in \mathbb{Z}_q} \left(\omega^{ik'} \mathcal{D}_1(k') \sum_{j \in \mathbb{Z}_q} \omega^{ij} \mathcal{D}_0(j) \right)$$
$$= \widehat{\mathcal{D}_0}(i) \cdot \widehat{\mathcal{D}_1}(i).$$

2. Let us prove the inversion.

$$\sum_{k \in \mathbb{Z}_q} \widehat{\mathcal{D}}(k) \omega^{-jk} = \sum_{k \in \mathbb{Z}_q} \left(\sum_{l \in \mathbb{Z}_q} \omega^{kl} \mathcal{D}(l) \right) \omega^{-jk}$$
$$= \sum_{l \in \mathbb{Z}_q} \mathcal{D}(l) \sum_{k \in \mathbb{Z}_q} \omega^{k(l-j)}$$
$$= q\mathcal{D}(j).$$

The last equation comes from the fact that
$$\sum_{k \in \mathbb{Z}_q} \omega^{km} = \begin{cases} q \text{ if } m = 0 \\ 0 \text{ else (as the sum of a geometric sequence)} \end{cases}$$

□

B Trivial reductions of Theorem 3

Remark 5. LWE self-reduction. In the case where $\mathsf{LWE\text{-}OD}^n_{\varphi_0,\dots,\varphi_{q-1}}(\boldsymbol{k})$ is exactly the $\mathsf{LWE}^n_\psi(\boldsymbol{k})$ distribution (i.e., $\varphi_0 = \cdots = \varphi_{q-1} = \psi$), we can use theorem conditions to prove that $\mathsf{LWE}^n_\psi(\boldsymbol{k})$ is at least as hard as $\mathsf{LWE}^n_\psi(\boldsymbol{k})$. This results in a trivial self-reduction. Indeed, plugging $\varphi_u = \psi$ for all u into the condition of Theorem 3, we get:

$$\sum_{\substack{k \in \mathbb{Z}_q \\ \widehat{\psi}(k) \neq 0}} \sum_{\ell=0}^{q-1} \frac{\varphi_{i-\ell}(\ell)}{\widehat{\psi}(k)} \omega^{k \cdot (\ell - j)} = \sum_{k=0}^{q-1} \sum_{\ell=0}^{q-1} \frac{\psi(\ell)}{\widehat{\psi}(k)} \omega^{k \cdot (\ell - j)}$$
$$= \sum_{k=0}^{q-1} \frac{\widehat{\psi}(k)}{\widehat{\psi}(k)} \omega^{-jk}$$
$$= \sum_{k=0}^{q-1} \omega^{-jk}$$
$$= \begin{cases} q \text{ if } j = 0, \\ 0 \text{ else.} \end{cases}$$

This allows to conclude that the condition is verified for any ψ and $\varphi_0 = \cdots = \varphi_{q-1} = \psi$.

Remark 6. Any LWE-OD instance can be unconditionally reduced from the LWE distribution with ψ being a Dirac function:

$$\psi(x) = \begin{cases} 1 \text{ if } x = 0, \\ 0 \text{ else.} \end{cases}$$

In that case, $\widehat{\psi}(k) = 1$ for any $k \in \mathbb{Z}_q$, validating the condition of Theorem 3. This reduction is however pointless as LWE with ψ being a Dirac has no error and can be solved with linear algebra. This is a purely formal reduction.

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